

Subtracting Integers: KCC Rule

Keep-Change-Change Rule

The Keep-Change-Change rule is a clever way to remember how to rewrite any subtraction problem as an addition problem.

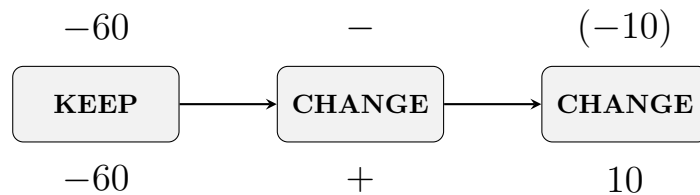
Procedure 1 (Keep-Change-Change Rule). Any subtraction problem can be rewritten as an equivalent addition problem using the steps below.

- (a) **Keep** the first number the same.
- (b) **Change** the operation of subtraction to addition.
- (c) **Change** the second number to its opposite.

Remember: Subtracting a number is the same as adding the opposite of that number.

Example 1. Use the Keep-Change-Change rule to find the difference $-60 - (-10)$.

Explanation. The diagram below demonstrates how to rewrite the subtraction problem $-60 - (-10)$ as an addition problem. Starting from the left, we **keep** the first number the same, **change** the operation of addition to subtraction, then **change** the second number to its opposite. Lastly, we add to find the result.



Alt text: A

horizontal flowchart showing the Keep-Change-Change rule for the expression negative sixty minus negative ten. Three boxes are arranged from left to right. The first box is labeled KEEP with the number negative sixty above and below it. An arrow points to the second box, labeled CHANGE, which shows a subtraction sign above it changing to an addition sign below it. A second arrow points to the third box, labeled CHANGE, which shows negative ten above it changing to positive ten below it.

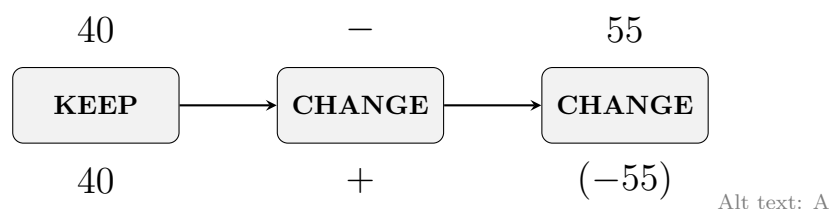
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$$\begin{aligned} -60 - (-10) &= -60 + 10 \\ &= \dots\dots \end{aligned}$$

Example 2. Use the Keep-Change-Change rule to find the difference $40 - 55$.

Explanation. The diagram below demonstrates how to rewrite the subtraction problem $40 - 55$ as an addition problem. Starting from the left, we **keep** the first number the same, **change** the operation of addition to subtraction, then **change** the second number to its opposite. Lastly, we add to find the result.



horizontal flowchart showing the Keep-Change-Change rule for the expression forty minus fifty five. Three boxes are arranged from left to right. The first box is labeled KEEP with the number forty above and below it. An arrow points to the second box, labeled CHANGE, which shows a subtraction sign above it changing to an addition sign below it. A second arrow points to the third box, labeled CHANGE, which shows negative fifty five above it changing to positive fifty five below it.

$$\begin{aligned} 40 - 55 &= 40 + (-55) \\ &= \dots\dots \end{aligned}$$

Problem 1 Use the **Keep-Change-Change** rule to rewrite the subtraction problem as an equivalent addition problem. Then find the answer.

$$-8 - (-10)$$

Multiple Choice:

- (a) $-8 + 10$
- (b) $-8 - 10$
- (c) $8 + 10$
- (d) $8 - 10$

Answer =

Problem 2 Use the **Keep-Change-Change** rule to rewrite the subtraction problem as an equivalent addition problem. Then find the answer.

$$200 - (-300)$$

Multiple Choice:

(a) $200 + 300$

(b) $200 - 300$

(c) $-200 + 300$

(d) $-200 - 300$

Answer =

Problem 3 Use the **Keep-Change-Change** rule to rewrite the subtraction problem as an equivalent addition problem. Then find the answer.

$$6 - 7$$

Multiple Choice:

(a) $6 + (-7)$

(b) $6 + 7$

(c) $-6 + (-7)$

(d) $-6 + 7$

Answer =

Problem 4 Use the **Keep-Change-Change** rule to rewrite the subtraction problem as an equivalent addition problem. Then find the answer.

$$-90 - (-20)$$

Multiple Choice:

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(a) $-90 + 20$

(b) $-90 - 20$

(c) $90 + 20$

(d) $90 - 20$

Answer =

Problem 5 *Is the operation of subtraction commutative? In other words, can we change the order of the numbers in a subtraction problem and still get the same result?*

Multiple Choice:

(a) Yes

(b) No

Remark 1. The rules for adding and subtracting integers apply to all real numbers. The integers, along with the fractions and decimals between them make up the set of **real numbers**. So, we can apply these rules to add and subtract fractions and decimals.